

L6a: Data-Driven Minimum-Variance Portfolios

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Risk. Trading involves risk and is generally not recommended for those with limited resources, experience, or a low risk tolerance. Past performance does not guarantee future success.

Responsibility. You are responsible for your investment decisions based on your financial circumstances, objectives, risk tolerance, and liquidity needs.

Objectives for Today

By the end of this lecture, you should be able to:

- **Formulate portfolio reward and risk:** derive matrix expressions for expected growth and variance, and explain diversification through covariance.
- **Construct minimum-variance portfolios:** compute GMV weights allowing shorts, identify the efficient frontier, and explain how long-only constraints, exact growth targets, and growth floors affect the solution.
- **Evaluate estimated allocations:** identify which estimated inputs each allocation depends on, and evaluate the resulting investment using wealth and NPV on prices reserved for testing.

Lectures and Examples

Lecture: Data-Driven Minimum-Variance Portfolios

Example 1: Portfolio Weights and Dirichlet Sampling

Draw long-only allocations, compare their estimated growth and risk, and follow selected buy-and-hold portfolios.

Example 2: Minimum-Variance Portfolios from Data

Estimate inputs from 2014–2024 data and trace the frontier. Compare both GMV portfolios, equal weights, and an index fund using 2025 prices.

Example 3: Portfolio Wealth with Multiple Asset GBM

Simulate correlated prices and buy-and-hold wealth, then estimate the probability of exceeding a target scaled NPV.

Concept Review: Portfolio Weights

How much of our money should we put into each asset? For M assets, w_i is the fraction of initial wealth invested in asset i :

$$\underbrace{\mathbf{w} = [w_1, \dots, w_M]^\top}_{\text{mix of assets}}, \quad \underbrace{w_i \geq 0}_{\text{long only}}, \quad \underbrace{\sum_{i=1}^M w_i = 1}_{\text{fully invested}}.$$

These constraints define the **allocation simplex**:

- Two assets give a line segment, and three assets give a triangle.
- Each corner invests the entire budget in one asset. Equal weights lie at the center.
- With initial wealth $W_0 > 0$, invest $w_i W_0$ dollars in asset i . Holding the purchased shares fixed allows the weights to change with prices.

We first sample feasible allocations, then develop an optimization problem to choose the weights.

Concept Review: Dirichlet Sampling

The [Dirichlet distribution](#) generates fully invested, long-only allocations. Draw $\mathbf{w} \sim \text{Dirichlet}(\alpha)$, with dimensionless $\alpha_i > 0$ and $\alpha_0 = \sum_i \alpha_i$. Each weight has:

$$\boxed{\mathbb{E}[w_i] = \frac{\alpha_i}{\alpha_0}}, \quad \boxed{\text{Var}(w_i) = \frac{\alpha_i(\alpha_0 - \alpha_i)}{\alpha_0^2(\alpha_0 + 1)}}.$$

- The ratios α_i/α_0 set the expected weights. Larger α_0 at fixed ratios reduces their spread across draws.
- With equal $\alpha_i = \alpha$: $\alpha < 1$ favors the boundary, $\alpha = 1$ samples uniformly, and large α clusters draws near equal weights.

Finding the minimum variance over all feasible weights requires optimization.

Example 1: [Portfolio Weights and Dirichlet Sampling](#)

Company Profile: AllianceBernstein

AllianceBernstein manages investments for institutional and individual clients.

- **Investment management:** equity, fixed-income, and multi-asset strategies, supported by quantitative research and data science.
- **Managing risk:** the [AB International Low Volatility Equity ETF](#) combines quantitative and fundamental research, emphasizing risk mitigation.
- **Artificial intelligence:** [Andrew Chin](#), Chief Artificial Intelligence Officer and Cornell alumnus, leads the firm's AI strategy. I know Andrew through the [AI in Finance short course](#).

Today's connection: how should we choose portfolio weights to balance expected growth and risk?

Explore: [Student opportunities](#) [Cornell Keynotes conversation with Andrew Chin](#)

Modern Portfolio Theory: Portfolio Reward

For M risky assets, w_i is the initial fraction invested in asset i , with $\sum_i w_i = 1$. Hold the weights fixed when calculating reward and risk. Let $\mathbf{g} = [g_1, \dots, g_M]^\top$ contain the asset growth rates over one observation interval. Their weighted average is:

$$g_p = \sum_{i=1}^M w_i g_i = \mathbf{w}^\top \mathbf{g}.$$

With fixed weights, **reward** is:

$$\mathbb{E}[g_p] = \sum_{i=1}^M w_i \mathbb{E}[g_i] = \mathbf{w}^\top \mathbb{E}[\mathbf{g}] \approx \mathbf{w}^\top \mathbf{g}'.$$

We estimate expected growth using the sample-mean vector $\mathbf{g}'$ (year⁻¹).

Expected growth is linear in the portfolio weights.

Modern Portfolio Theory: Portfolio Risk

We measure **risk** by $\text{Var}(g_p)$. With fixed weights and covariance matrix $\Sigma_g = \text{Cov}(\mathbf{g})$:

$$\begin{aligned}\text{Var}(g_p) &= \text{Cov}\left(\sum_i w_i g_i, \sum_j w_j g_j\right) \\ &= \sum_i \sum_j w_i w_j \text{Cov}(g_i, g_j) = \mathbf{w}^\top \Sigma_g \mathbf{w}.\end{aligned}$$

Individual variances (diagonal entries) and pairwise covariances (off-diagonal entries) both contribute to risk. From data, use $\hat{\Sigma}_g$.

Variance has units year^{-2} . Its square root, $\sigma_{g,p} = \sqrt{\text{Var}(g_p)}$, is the growth-rate standard deviation (year^{-1}).

To see how covariance can reduce risk, consider a two-asset portfolio.

Two-Asset Diversification

Invest fractions w and $1 - w$, with $0 < w < 1$, in two assets with positive standard deviations $\sigma_{g,1}$ and $\sigma_{g,2}$ and correlation ρ_{12} :

$$\text{Var}(g_p) = w^2\sigma_{g,1}^2 + (1 - w)^2\sigma_{g,2}^2 + 2w(1 - w) \underbrace{\rho_{12}\sigma_{g,1}\sigma_{g,2}}_{\text{Cov}(g_1, g_2)}.$$

Correlation determines what mixing achieves:

- At $\rho_{12} = 1$, $\sigma_{g,p} = w\sigma_{g,1} + (1 - w)\sigma_{g,2}$ is the weighted average.
- For $\rho_{12} < 1$, it falls below that weighted average.
- If $\sigma_{g,1} \leq \sigma_{g,2}$ and $\rho_{12} < \sigma_{g,1}/\sigma_{g,2}$, some mix has less variance than either asset.

The Global Minimum-Variance Portfolio

The **global minimum-variance (GMV) portfolio** has the smallest growth-rate variance among fully invested portfolios. With shorts allowed, some weights can be negative. For M risky assets with positive-definite covariance Σ_g and vector of ones $\mathbf{1}$, solve:

$$\begin{array}{ll} \underset{\mathbf{w} \in \mathbb{R}^M}{\text{minimize}} & \frac{1}{2} \mathbf{w}^\top \Sigma_g \mathbf{w} \quad \text{minimize growth-rate variance} \\ \text{subject to} & \mathbf{1}^\top \mathbf{w} = 1 \quad \text{weights sum to one.} \end{array}$$

The unique GMV weights and minimum variance are:

$$\mathbf{w}_{\text{GMV}} = \frac{\Sigma_g^{-1} \mathbf{1}}{\mathbf{1}^\top \Sigma_g^{-1} \mathbf{1}}, \quad \sigma_{g,\text{GMV}}^2 = \frac{1}{\mathbf{1}^\top \Sigma_g^{-1} \mathbf{1}}.$$

No mean-growth estimate is needed because there is no growth target.

Deriving the GMV Weights

Introduce a multiplier λ for the budget constraint. The factor $1/2$ simplifies the derivative without changing the minimizing weights:

$$\mathcal{L}(\mathbf{w}, \lambda) = \frac{1}{2} \mathbf{w}^\top \Sigma_g \mathbf{w} - \lambda (\mathbf{1}^\top \mathbf{w} - 1).$$

- **Stationarity:** $\Sigma_g \mathbf{w} - \lambda \mathbf{1} = \mathbf{0}$, so $\mathbf{w} = \lambda \Sigma_g^{-1} \mathbf{1}$.
- **Budget:** $1 = \lambda \mathbf{1}^\top \Sigma_g^{-1} \mathbf{1}$, giving $\lambda = (\mathbf{1}^\top \Sigma_g^{-1} \mathbf{1})^{-1}$.
- **Variance:** use $\Sigma_g \mathbf{w} = \lambda \mathbf{1}$ and $\mathbf{w}^\top \mathbf{1} = 1$:

$$\sigma_{g,\text{GMV}}^2 = \mathbf{w}^\top \Sigma_g \mathbf{w} = \lambda \mathbf{w}^\top \mathbf{1} = \lambda = \frac{1}{\mathbf{1}^\top \Sigma_g^{-1} \mathbf{1}}.$$

Positive definiteness makes the objective strictly convex, so these weights give the unique minimum.

Derivation: [Complete GMV derivation companion](#)

What Does a Negative Weight Mean?

For two assets with $\sigma_{g,1} \leq \sigma_{g,2}$, the GMV portfolio shorts the riskier asset when:

$$\rho_{12} > \frac{\sigma_{g,1}}{\sigma_{g,2}}.$$

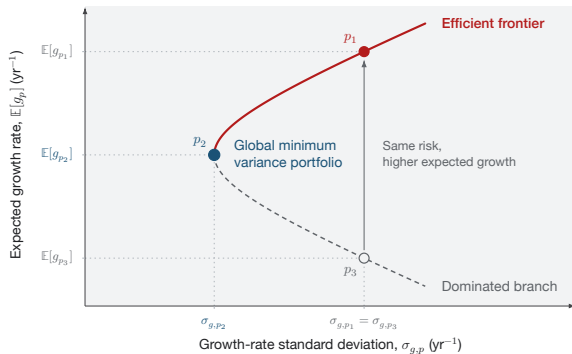
With 1,000 USD and weights $w_1 = 1.2$ and $w_2 = -0.2$:

- Borrow and sell 200 USD of asset 2, then buy 1,200 USD of asset 1.
- We must repurchase the borrowed shares, so the short gains when the price of the borrowed shares falls and loses when the price rises.

We ignore borrowing fees, collateral, and position limits.

The Minimum-Variance Frontier

Unlike the long-only samples in Example 1, we now allow short positions:



- The curve gives the smallest variance at each growth rate. Every feasible portfolio lies on or to its right.
- p_1 has higher growth than p_3 at the same risk, so p_3 and the lower branch are dominated.
- The upper branch, starting at the GMV portfolio p_2 , is the **efficient frontier**.

Efficient: No feasible portfolio offers at least as much growth and no more variance, with a strict improvement in either.

The Target-Growth Problem

How do we calculate the minimum-variance frontier? Given sample means $\mathbf{g}'$ and a target $g_\star$ (year⁻¹) for estimated expected growth, solve:

minimize	$\mathbf{w}^\top \Sigma_g \mathbf{w}$	minimize growth-rate variance
subject to	$(\mathbf{g}')^\top \mathbf{w} = g_\star$	meet the exact growth target
	$\mathbf{1}^\top \mathbf{w} = 1$	weights sum to one.

Short positions are allowed, and Σ_g is positive definite.

The sample means must not all be equal, since otherwise every portfolio has the same estimated expected growth.

Optimal Weights at an Exact Growth Target

For $(\mathbf{g}')^\top \mathbf{w} = g_\star$ with short positions allowed, define:

$$\begin{aligned} a &= \mathbf{1}^\top \Sigma_g^{-1} \mathbf{1}, & b &= \mathbf{1}^\top \Sigma_g^{-1} \mathbf{g}', \\ c &= (\mathbf{g}')^\top \Sigma_g^{-1} \mathbf{g}', & d &= ac - b^2 > 0. \end{aligned}$$

The unique minimizing weights are:

$$\mathbf{w}_\star(g_\star) = \frac{c - bg_\star}{d} \Sigma_g^{-1} \mathbf{1} + \frac{ag_\star - b}{d} \Sigma_g^{-1} \mathbf{g}'.$$

These weights sum to one and meet the growth target exactly.

When $g_\star = b/a$, the second term vanishes and we recover the GMV weights, $\mathbf{w}_{\text{GMV}} = \Sigma_g^{-1} \mathbf{1} / a$.

Derivation: [Complete target-growth derivation companion](#)

Minimum Variance at a Growth Target

Using the same coefficients, the smallest variance at target g_* , in year^{-2} , is:

$$\sigma_g^2(g_*) = \frac{ag_*^2 - 2bg_* + c}{d} = \underbrace{\frac{1}{a}}_{\text{GMV variance}} + \underbrace{\frac{a}{d} \left(g_* - \frac{b}{a}\right)^2}_{\text{extra variance for the target}}.$$

Here, b/a is the expected growth rate of the GMV portfolio p_2 . Varying g_* traces the **minimum-variance frontier**, and $g_* \geq b/a$ gives the efficient frontier.

The Long-Only Problem

Many investors and funds cannot take short positions, and a growth requirement is usually a minimum. With sample means $\mathbf{g}'$, estimated covariance $\hat{\Sigma}_g$, and growth floor g_* , solve:

minimize	$\mathbf{w}^\top \hat{\Sigma}_g \mathbf{w}$	minimize estimated variance
subject to	$(\mathbf{g}')^\top \mathbf{w} \geq g_*$	meet or exceed the floor
	$\mathbf{1}^\top \mathbf{w} = 1$	weights sum to one
	$0 \leq w_i \leq 1, \quad i = 1, \dots, M$	exclude short positions.

Positive definiteness makes the solution unique. If the short-allowed GMV weights are feasible, they stay optimal. Otherwise, we solve numerically.

The Growth Floor

The effect of g_* depends on the estimated expected growth of the **long-only GMV portfolio**, found without a growth constraint:

- **At or below it:** the long-only GMV portfolio meets the floor and stays optimal.
- **Above it:** a feasible floor binds, so the optimal portfolio meets the target exactly. Raising the floor traces the efficient frontier, never the dominated branch. Growth averages the sample means, so $\max_i g'_i$ is the highest feasible floor.
- **Compared with shorts allowed:** with the same inputs, the long-only frontier lies on or to the right of the short-allowed frontier. The two touch where the short-allowed weights are nonnegative.

The data example solves both problems and traces the long-only frontier.

Example 2: [Minimum-Variance Portfolios from Data](#)

Estimated Inputs

The GMV weights use only the estimated covariance $\hat{\Sigma}_g$. A growth target or floor also uses the mean growth rates $\mathbf{g}'$, which are harder to pin down:

- For AMD (2014–2024), the sample mean gives 0.3120 per year, while L4b's log-price regression gives 0.4397 per year.
- The optimizer treats its inputs as exact, so it favors asset combinations whose variance is underestimated or whose mean growth is overestimated.

We therefore evaluate allocations on prices that were not used to choose them.

Portfolio Wealth and Scaled NPV

Buy $n_i = w_i W_0 / S_i(0)$ shares, with $w_i \geq 0$ and $\sum_i w_i = 1$, and hold them fixed. For a sale after T years at constant benchmark growth rate g_y :

$$W_t = W_0 \sum_{i=1}^M w_i \frac{S_i(t)}{S_i(0)} \quad \text{portfolio value in dollars}$$
$$\rho_T = \frac{\text{NPV}(g_y, T)}{W_0} = \frac{W_T}{W_0} e^{-g_y T} - 1 \quad \text{discounted gain per dollar invested.}$$

The sign of ρ_T shows whether the investment beats or trails the benchmark. We allow fractional shares and omit dividends and trading costs.

Example 2 applies W_t to the held-out 2025 prices. Example 3 simulates wealth with the fitted model to estimate $\mathbb{P}(\rho_T > \rho_\star)$.

Example 3: Portfolio Wealth with Multiple Asset GBM

Optional Advanced Material

This example develops the frontier geometry further.

Advanced: Frontier Geometry and the Two-Fund Theorem

How can two risky frontier portfolios generate the rest of the curve? Derive the weights and vary the mix of two fixed funds. Compare the frontier with a 50% limit on each position's magnitude and with long-only weights.

Summary

- **Portfolio reward and risk:** we expressed expected growth and variance using asset growth rates and portfolio weights. Covariance determines how combining assets changes risk and creates diversification.
- **Minimum-variance portfolios:** we obtained the GMV weights and identified the efficient frontier. With long-only constraints, sweeping a growth floor returns the long-only GMV portfolio and the efficient frontier.
- **Estimated inputs and realized performance:** we connected the estimated inputs to the weights and evaluated wealth and NPV on prices reserved for testing. An allocation optimized on historical estimates need not perform best.

In L6b, we use a single index model to estimate the inputs and add a risk-free asset.

Next: model asset growth using a shared market factor and firm-specific risk.